

Five Most Interesting Things I Learned from These Computer Vision PDFs

While going through the computer vision material, I found several concepts that were particularly interesting, especially from the perspective of healthcare applications. Instead of focusing only on algorithms, these topics helped me understand how humans see the world and how computer vision systems try to replicate that process.

1. Human Vision Works Like a Multi-Stage Processing System

One thing that surprised me was that our brain does not simply "see" an image. Different parts of the visual system process different kinds of information such as edges, motion, color, depth, and object recognition. This is very similar to how modern computer vision models work. Before making a final prediction, AI systems first extract low-level features and gradually build an understanding of what is present in the image.

For healthcare applications, this explains why feature extraction is so important when analyzing X-rays, CT scans, or MRI images.

2. The Fovea Shows That Not Every Pixel Is Equally Important

The fovea is a tiny region of the retina responsible for our sharpest vision. Humans constantly move their eyes so that important details fall onto this area. I found this fascinating because modern AI models use a very similar idea through attention mechanisms.

In medical imaging, diseases often occupy only a small region of an image. An AI system that can focus on suspicious areas rather than processing every pixel equally can be more efficient and accurate.

3. Color Is Actually a Brain Interpretation

Before reading these slides, I assumed color was a direct property of objects. The material explains that color is actually created by our brain based on signals received from three types of cone cells. Different light patterns can even appear identical to us.

This is particularly relevant in healthcare because medical images collected from different devices may look slightly different even when they represent the same tissue. Understanding this helps explain why image normalization is an important step before training AI models.

4. Segmentation Is More Useful Than Simple Classification

Many people think computer vision is only about identifying whether something is present or not. However, segmentation goes a step further by identifying the exact location and boundaries of an object.

In healthcare, this is extremely valuable. Doctors often need to know where a tumor is located and how large it is, not just whether it exists. This made me realize that segmentation can provide much more practical clinical information than a simple yes-or-no prediction.

5. Small Changes in Image Processing Can Affect Results

The discussion on image resizing and interpolation seemed simple at first, but it highlighted an important issue. When images are resized incorrectly, important details can be lost or distorted.

For medical images, even a tiny feature may indicate a disease. A small nodule in a lung scan or a subtle abnormality in an X-ray could disappear if preprocessing is not handled carefully. This showed me that data preparation can be just as important as the AI model itself.